

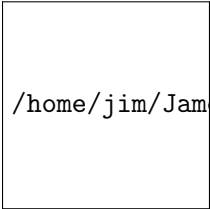
Section 2.3: Introduction to Combinatorics

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Wednesday 27th August, 2014



`/home/jim/JamesAshe/MHU/images/MarsHillU`

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Fundamental Principle of Counting

Example 2.3.1.

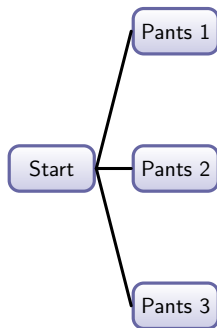
How many different outfits can worn from 3 pants, 2 shirts, and 1 pair of shoes?

Start

Fundamental Principle of Counting

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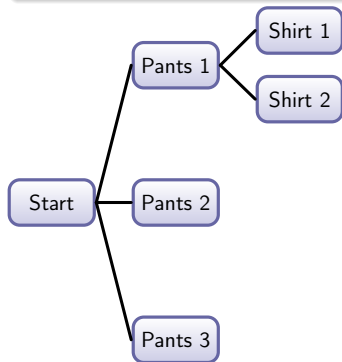
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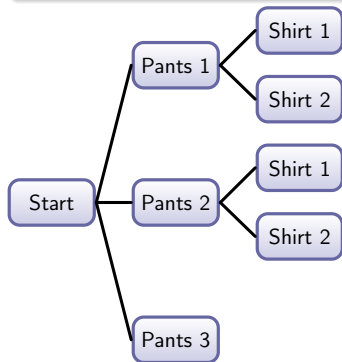
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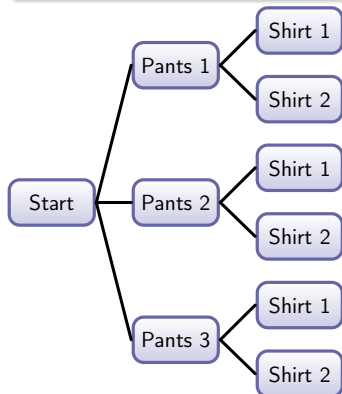
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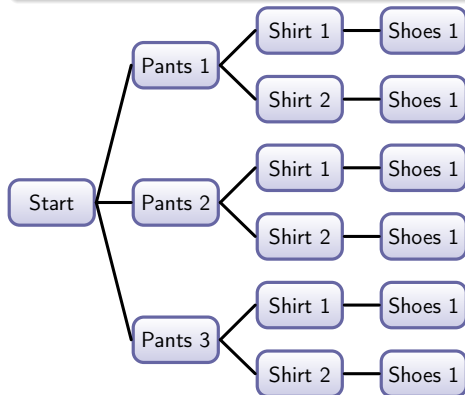
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1. Write blanks for each event/decision.

Example 2.3.2.

How many different outfits can worn from 3 pants, 4 shirts, and 10 pairs of shoes?

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How many different outfits can worn from 3 pants, 4 shirts, and 10 pairs of shoes?

$$\underline{3} \times \underline{4} \times \underline{10}$$

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Example 2.3.2.

How many different outfits can worn from 3 pants, 4 shirts, and 10 pairs of shoes?

$$\underline{3} \times \underline{4} \times \underline{10} = 120$$

Fundamental Principle of Counting

Example 2.3.3.

Three coins are tossed.

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- a. Use the Fundamental Principle of Counting to determine how many different outcomes are possible.

Fundamental Principle of Counting

Example 2.3.3.

Three coins are tossed.

- a. Use the Fundamental Principle of Counting to determine how many different outcomes are possible.
- b. Construct a tree diagram to list all possible outcomes.

How different ways can we arrange 3 things?

Factorials

How different ways can we arrange 3 things?

How different ways can we arrange 4 things?

Factorials

How different ways can we arrange 3 things?

How different ways can we arrange 4 things?

How different ways can we arrange 5 things?

Factorials

How different ways can we arrange 3 things?

How different ways can we arrange 4 things?

How different ways can we arrange 5 things?

How different ways can we arrange n things?

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Let n be a positive integer, then n *factorial* is

Special case: $0! = 1$.

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ex. $\frac{7!}{5!}$

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$$= \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

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$$\frac{7!}{3! \cdot 5!}$$

$$= \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(3 \cdot 2 \cdot 1)(5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)}$$

$$= \frac{7 \cdot 6}{6} = 7$$

Problem 2.3.1. A sandwich shop has 2 different breads, 8 meats, 5 cheeses, and 4 sauces.

How many different sandwiches are possible?

Problem 2.3.2. A password has 2 letters followed by 4 digits (0-9).

How many different passwords are possible if,

- a. Letters and digits can be repeated.

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Problem 2.3.3. Find the indicated value

$$\frac{90!}{89!}$$

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Problem 2.3.4. Find the indicated value

$$\frac{90!}{88! \cdot 3!}$$